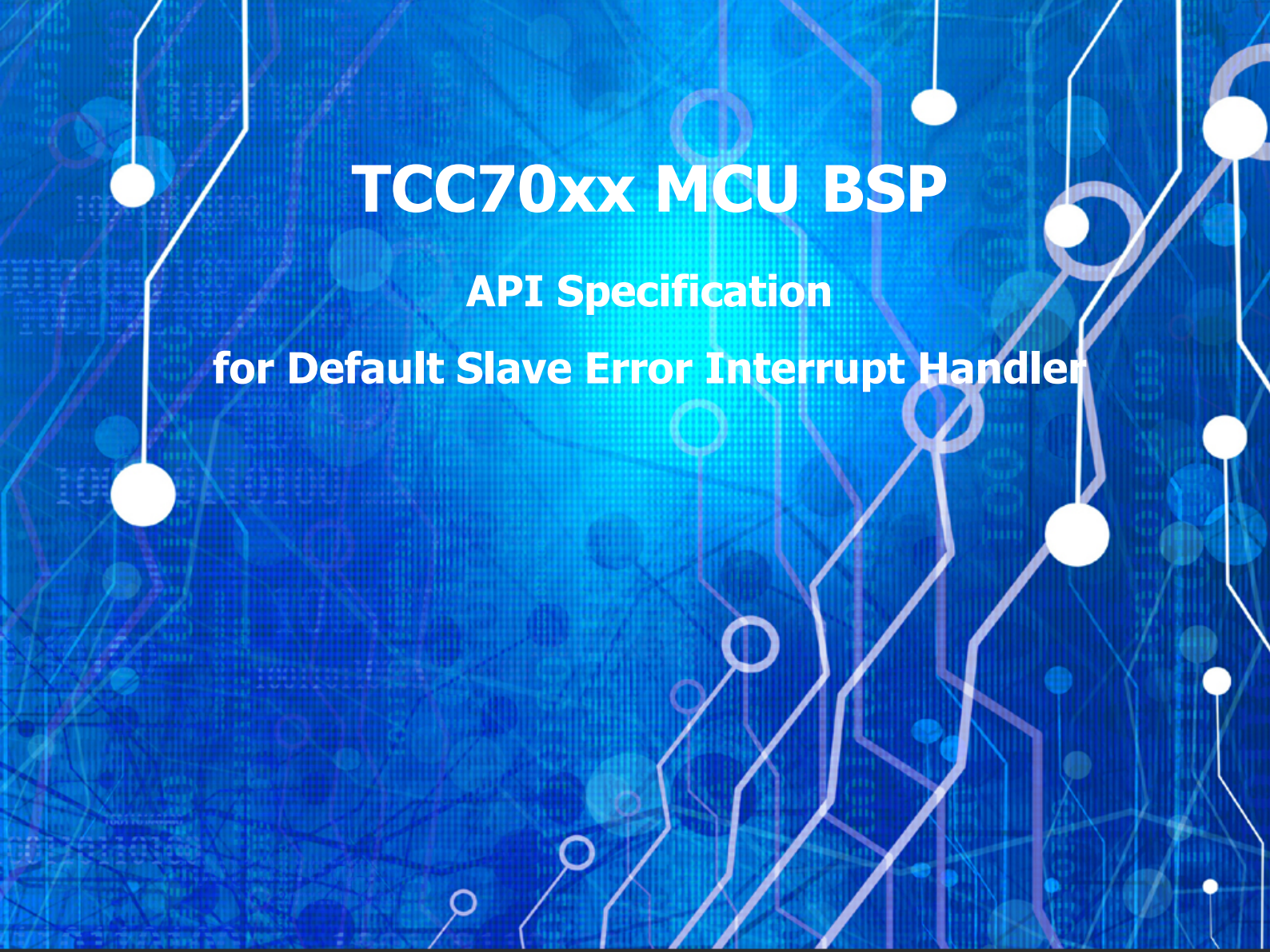




TCC70xx MCU BSP

API Specification

for Default Slave Error Interrupt Handler

Rev. 0.10 [A]

2021-09-03

Preliminary version

※ The information in this document is subject to change without notice and should not be construed as a commitment by Telechips Inc.

Kindly visit www.telechips.com for more information.

© 2021 Telechips Inc. All rights reserved.

TABLE OF CONTENTS

Contents

1	Introduction	3
2	Terms and Abbreviations	3
3	Source Files.....	4
3.1	Location of Source Files.....	4
3.2	Simple Description of Source Files.....	4
4	Constants.....	5
4.1	Default Slave Error Configuration Flag Values	5
4.2	Register Duplication Check Configuration Values.....	6
5	Structures	7
5.1	DSEReg_t	7
6	APIs	9
6.1	DSE_Init	9
6.2	DSE_Deinit.....	10
6.3	DSE_WriteLock.....	11
6.4	DSE_GetGrpSts	12
7	How to use DSE Driver	13
7.1	Example: Using DSE Register Redundancy Check	13
8	References	15
9	Revision History	16
	Rev. 0.10: 2021-09-03	16

Figures

Figure 3.1 Location of Source Files	4
---	---

Tables

Table 2.1 Terms and Abbreviation	3
--	---

1 INTRODUCTION

This document describes the default slave error interrupt handler device driver (hereinafter referred to as DSE driver) of TCC70xx MCU BSP. For more detailed information on default slave error interrupt handler in the TCC70xx MCU Sub-system, refer to Chapter 29 Default Slave Error Interrupt Handler in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

DSE	Default Slave Error
------------	---------------------

3 SOURCE FILES

3.1 Location of Source Files

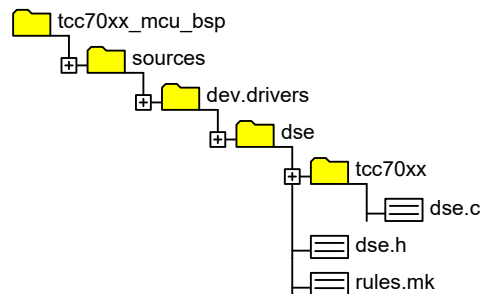


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `dse.c`
 - Driver source of DSE driver
- `dse.h`
 - API header of DSE driver
- `rules.mk`
 - Rules used for build

4 CONSTANTS

4.1 Default Slave Error Configuration Flag Values

These constants are used to determine which DSE module will be enabled.

Declaration

```
#define DSE_INTERCON      0x00000001u
#define DES_SEL_ALL       (DSE_INTERCON)
#define DES_SEL_MASK      DES_SEL_ALL
#define DSE_INT_MODE      0x00000000u  /* Default Slave Error interrupt mode */
#define DSE_RES_MODE      0x10000000u  /* Default Slave Error response mode */
#define DSE_RES_INT_MODE  0x20000000u  /* Default Slave Error int & res mode */
#define DSE_MODE_MASK     0x30000000u
```

Description

Value	Description
<i>DSE_INTERCON</i>	<i>Enable Peripheral AHB Interconnector's default slave error handler.</i>
<i>DES_SEL_ALL</i>	<i>Enable all of the default slave error handlers.</i>
<i>DES_SEL_MASK</i>	<i>Mask value for selecting module of the default slave error handler in the <i>DSE_Init</i> input parameter.</i>
<i>DSE_INT_MODE</i>	<i>Enable Error IRQ of the default slave error handler.</i>
<i>DSE_RES_MODE</i>	<i>Enable Error Response of the default slave error handler.</i>
<i>DSE_RES_INT_MODE</i>	<i>Enable Error IRQ and Error Response of the default slave error handler.</i>
<i>DSE_MODE_MASK</i>	<i>Mask value for selecting mode of the default slave error handler in the <i>DSE_Init</i> input parameter.</i>

Remark

In response mode (*DSE_RES_MODE* & *DSE_RES_INT_MODE*), when the CPU receives an error response, data abort occurs in the CPU. In IRQ mode (*DSE_INT_MODE* & *DSE_RES_INT_MODE*), the CPU receives the IRQ from the default slave error IRQ handler.

4.2 Register Duplication Check Configuration Values

These constants are used to check register duplication.

Declaration

```
#define DSE_SM_IRQ_MASK          0x2u
#define DSE_SM_IRQ_EN           0x4u
#define DSE_SM_SLV_CFG          0x8u
#define DSE_SM_CFG_WR_PW        0x10000u
#define DSE_SM_CFG_WR_LOCK      0x20000u
#define DSE_SM_SF_CHK_GRP_SEL    (DSE_SM_IRQ_MASK | DSE_SM_IRQ_EN | DSE_SM_SLV_CFG
                                   | DSE_SM_CFG_WR_PW | DSE_SM_CFG_WR_LOCK)
```

Description

Value	Description
<i>DSE_SM_IRQ_MASK</i>	<i>Bit field value for enabling soft fault status check or soft fault check of DSE_IRQ_MASK register</i>
<i>DSE_SM_IRQ_EN</i>	<i>Bit field value for enabling soft fault status check or soft fault check of DSE_IRQ_EN register</i>
<i>DSE_SM_SLV_CFG</i>	<i>Bit field value for enabling soft fault status check or soft fault check of DEF_SLV_CFG register</i>
<i>DSE_SM_CFG_WR_PW</i>	<i>Bit field value for enabling soft fault status check or soft fault check of DSE_IH_CFG_WR_PW register</i>
<i>DSE_SM_CFG_WR_LOCK</i>	<i>Bit field value for enabling soft fault status check or soft fault check of DSE_IH_CFG_WR_PW register</i>
<i>DSE_SM_SF_CHK_GRP_SEL</i>	<i>Bit field value for enabling soft fault check</i>

Remark

For the description of DSE_IRQ_MASK, DSE_IRQ_EN, DEF_SLV_CFG, DSE_IH_CFG_WR_PW, DSE_IH_CFG_WR_LOCK register, refer to Chapter 29 Default Slave Error Interrupt Handler in the “*TCC70xx Full Specification*” [1].

5 STRUCTURES

5.1 DSEReg_t

The DSE register is accessed by mapping to the following structure:

Declaration

```
typedef struct DSEReg
{
    uint32 rDSE_IRQ_STATUS;
    uint32 rDSE_IRQ_MASK;
    uint32 rDSE_IRQ_EN;
    uint32 rDEF_SLV_CFG;
    uint32 rDEF_SLV0_ADDR;
    uint32 rDEF_RESERVED[11];
    uint32 rDSE_IH_CFG_WR_PW;
    uint32 rDSE_IH_CFG_WR_LOCK;
    uint32 rSF_CHK_GRP_EN;
    uint32 rSF_CHK_GRP_STS;
} DSEReg_t;
```

Description

Value	Description
<i>rDSE_IRQ_STATUS</i>	Default Slave Error Interrupt Status
<i>rDSE_IRQ_MASK</i>	Default Slave Error Interrupt Mask
<i>rDSE_IRQ_EN</i>	Default Slave Error Interrupt Enable
<i>rDEF_SLV_CFG</i>	Default Slave Interrupt/Error Response Configuration
<i>rDEF_SLV0_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_INTERCON</i> , <i>rDEF_SLV0_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_RESERVED</i>	<u>Not</u> used
<i>rDSE_IH_CFG_WR_PW</i>	If you want to write to <i>rDSE_IH_CFG_WR_LOCK</i> or <i>rSF_CHK_GRP_EN</i> , you should first write a predefined password value (0x5AFEACE5) to <i>rDSE_IH_CFG_WR_PW</i> . If you successfully write the predefined password value (0x5AFEACE5) to <i>rDSE_IH_CFG_WR_PW</i> , <i>rDSE_IH_CFG_WR_PW</i> is read as 1.
<i>rDSE_IH_CFG_WR_LOCK</i>	Default Slave Error Interrupt Handler Configuration Write Lock. When <i>rDSE_IH_CFG_WR_LOCK</i> is written as 1, the following values <u>cannot</u> be written. ■ <i>rDSE_IRQ_MASK</i> ■ <i>rDSE_IRQ_EN</i> ■ <i>rDEF_SLV_CFG</i>
<i>rSF_CHK_GRP_EN</i>	Default Slave Error Interrupt Handler Configuration Soft Fault Check Group Enable. Refer to Chapter 4.2.

<i>Value</i>	<i>Description</i>
<i>rSF_CHK_GRP_STS</i>	<i>Default Slave Error Interrupt Handler Configuration Soft Fault Check Group Status. If an error occurs in the register set in rSF_CHK_GRP_EN, you can read rSF_CHK_GRP_STS to identify which register caused the error.</i>

Remark

None

6 APIs

6.1 DSE_Init

Initializes a DSE interrupt.

Declaration

```
SALRetCode_t DSE_Init  
(  
    uint32 uiModeFlag  
);
```

Parameters

Value	Description
uiModeFlag	[in] Refer to Chapter 4.1.

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

6.2 DSE_Deinit

De-initializes a DSE interrupt.

Declaration

```
void DSE_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 DSE_WriteLock

Enables or disables write protection of the DSE register.

Declaration

```
SALRetCode_t DSE_WriteLock  
(  
    uint32 uiLock  
);
```

Parameters

Value	Description
<i>uiLock</i>	<i>[in] Write Protection 1: Enable, 0: Disable.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.4 DSE_GetGrpSts

Gets the *SF_CHK_GRP_STS* register value. Refer to Chapter 5.1 DSEReg_t.

Declaration

```
uint32 DSE_GetGrpSts
(
    void
);
```

Parameters

None

Return

Value	Description
<i>DSE_SM_IRQ_MASK</i>	Unintended changes occur in the <i>DSE_IRQ_MASK</i> register.
<i>DSE_SM_IRQ_EN</i>	Unintended changes occur in the <i>DSE_IRQ_EN</i> register.
<i>DSE_SM_SLV_CFG</i>	Unintended changes occur in the <i>DEF_SLV_CFG</i> register.
<i>DSE_SM_CFG_WR_PW</i>	Unintended changes occur in the <i>DSE_IH_CFG_WR_PW</i> register.
<i>DSE_SM_CFG_WR_LOCK</i>	Unintended changes occur in the <i>DSE_IH_CFG_WR_LOCK</i> register.

Remark

None

7 HOW TO USE DSE DRIVER

You can use the DSE driver by calling the *DSE_Init* function. The currently released code is designed to output a debug message when a default slave error interrupt occurs.

7.1 Example: Using DSE Register Redundancy Check

The following example shows how to enable redundancy checking of DSE's registers: mission register and duplicated register. For detailed description on register setting, refer to Chapter 30 System Safety Mechanism (SM) Control Module in the "*TCC70xx Full Specification*"[1].

```
void DSE_Test_SoftFaultCheck (void)
{
    #if defined(FMU_DRIVER)
        uint32 uiRdata    = 0UL;
        uint32 uiErr      = 0UL;
        uint32 uiFmuArg[2];
        uint32 uiI        = 0UL;

        uiRdata          = 0UL;
        uiErr             = 0UL;
        uiI               = 0UL;
        uiFmuArg[0]       = 0UL;
        uiFmuArg[1]       = 0UL;

        DSE_Init(DES_SEL_ALL);
        (void)DSE_Init(DES_SEL_ALL);
        (void)FMU_IsrHandler((FMUFaultid_t)FMU_ID_SYS_SM_CFG, (FMUSeverityLevelType_t)FMU_SVL_LOW,
            (FMUIntFnctPtr)&DSE_TEST_Fmu_IRQ_ISR, (void *)uiFmuArg);
        (void)FMU_Set((FMUFaultid_t)FMU_ID_SYS_SM_CFG);
        (void)FMU_IsrClr((FMUFaultid_t)FMU_ID_SYS_SM_CFG);

        if(uiErr == 0UL)
        {
            if(DSE_WriteLock(0UL) == SAL_RET_SUCCESS)
            {
                if(uiRdata != 1UL)
                {
                    mcu_printf(" Fail.. DSETEST_SM_SF_CTRL_PW set\n");
                    uiErr |= (0x10UL);
                }
                else
                {
                    uiRdata = SAL_ReadReg(DSETEST_SM_SF_CTRL_CFG);
                    /* for write timeout to 0xffff and dse fault en, test mod */
                    SAL_WriteReg(uiRdata | 0xFFF0004UL, DSETEST_SM_SF_CTRL_CFG);
                    mcu_printf(" SF_CTRL_CFG set\n");
                }
                ...
            }
        }
    }
}
```

```
static void DSE_TEST_Fmu_IRQ_ISR(void *pArg)
{
    uint32 uiRdata    = 0UL;
    uint32 uiReadSts  = 0UL;

    (void)pArg;
    uiReadSts = SAL_ReadReg(DSETEST_SM_SF_CTRL_STS);
    printf("FMU interrupt ... SF_CTRL_STS 0x%x\n", uiReadSts);

    if(uiReadSts == 0x4UL)
    {
        uiRdata = DSE_GetGrpSts();

        if((uiRdata & DSE_SM_IRQ_MASK) == DSE_SM_IRQ_MASK)
        {
            printf("DSE_IRQ_MASK error\n");
        }
        if((uiRdata & DSE_SM_IRQ_EN) == DSE_SM_IRQ_EN)
        {
            printf("DSE_SM_IRQ_EN error\n");
        }
        if((uiRdata & DSE_SM_SLV_CFG) == DSE_SM_SLV_CFG)
        {
            printf("DSE_SM_SLV_CFG error\n");
        }
        if((uiRdata & DSE_SM_CFG_WR_PW) == DSE_SM_CFG_WR_PW)
        {
            printf("DSE_SM_CFG_WR_PW error\n");
        }
        if((uiRdata & DSE_SM_CFG_WR_LOCK) == DSE_SM_CFG_WR_LOCK)
        {
            printf("DSE_SM_CFG_WR_LOCK error\n");
        }
    }
}
```

8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

DISCLAIMER

All information and data contained in this material are without any commitment, are not to be considered as an offer for conclusion of a contract, nor shall they be construed as to create any liability. Any new issue of this material invalidates previous issues. Product availability and delivery are exclusively subject to our respective order confirmation form; the same applies to orders based on development samples delivered. By this publication, Telechips, Inc. does not assume responsibility for patent infringements or other rights of third parties that may result from its use.

Further, Telechips, Inc. reserves the right to revise this publication and to make changes to its content, at any time, without obligation to notify any person or entity of such revisions or changes.

No part of this publication may be reproduced, photocopied, stored on a retrieval system, or transmitted without the express written consent of Telechips, Inc.

This product is designed for general purpose, and accordingly customer be responsible for all or any of intellectual property licenses required for actual application. Telechips, Inc. does not provide any indemnification for any intellectual properties owned by third party.

Telechips, Inc. can not ensure that this application is the proper and sufficient one for any other purposes but the one explicitly expressed herein. Telechips, Inc. is not responsible for any special, indirect, incidental or consequential damage or loss whatsoever resulting from the use of this application for other purposes.

COPYRIGHT STATEMENT

Copyright in the material provided by Telechips, Inc. is owned by Telechips unless otherwise noted.

For reproduction or use of Telechips' copyright material, permission should be sought from Telechips. That permission, if given, will be subject to conditions that Telechips' name should be included and interest in the material should be acknowledged when the material is reproduced or quoted, either in whole or in part. You must not copy, adapt, publish, distribute or commercialize any contents contained in the material in any manner without the written permission of Telechips. Trade marks used in Telechips' copyright material are the property of Telechips.

Important Notice

This material may include technology owned by the 3rd party licensor and the technology may be subject to its associated licenses. It is solely customer's responsibility to identify and comply with such licenses. No other licenses are granted or implied by Telechips with making available this material.

For customers who use licensed Codec ICs and/or licensed codec firmware of mp3:

"Supply of this product does not convey a license nor imply any right to distribute content created with this product in revenue-generating broadcast systems (terrestrial. Satellite, cable and/or other distribution channels), streaming applications(via internet, intranets and/or other networks), other content distribution systems(pay-audio or audio-on-demand applications and the like) or on physical media(compact discs, digital versatile discs, semiconductor chips, hard drives, memory cards and the like). An independent license for such use is required. For details, please visit <http://mp3licensing.com>".

For customers who use other firmware of mp3:

"Supply of this product does not convey a license under the relevant intellectual property of Thomson and/or Fraunhofer Gesellschaft nor imply any right to use this product in any finished end user or ready-to-use final product. An independent license for such use is required. For details, please visit <http://mp3licensing.com>".

For customers who use Digital Wave DRA solution:

"Supply of this implementation of DRA technology does not convey a license nor imply any right to this implementation in any finished end-user or ready-to-use terminal product. An independent license for such use is required."

For customers who use DTS technology:

"This product made under license to certain U.S. patents and/or foreign counterparts."
"© 1996 – 2011 DTS, Inc. All rights reserved."

For customers who use Dolby technology:

"Supply of this Implementation of Dolby technology does not convey a license nor imply a right under any patent, or any other industrial or intellectual property right of Dolby Laboratories, to use this Implementation in any finished end-user or ready-to-use final product. It is hereby notified that a license for such use is required from Dolby Laboratories."

For customers who use Google technology:

"Copyright © 2013 Google Inc. All rights reserved."

For customers who use MS technology:

"This product is subject to certain intellectual property rights of Microsoft and cannot be used or distributed further without the appropriate license(s) from Microsoft."